Clinical Study Summary Report (CSSR) Document Status: Final | **Version:** 1.0 | **Date:** 04-Apr-2026

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1. Administrative & Study Identification

Full Study Title: A Multicentre, Randomised, Double-Blind, Parallel-Group, Placebo-Controlled, Phase 3, Efficacy and Safety Study of Tezepelumab in 5 to < 12 Year Old Children With Severe Uncontrolled Asthma (HORIZON) **Brief Study Title:** A Study to Investigate the Efficacy and Safety of Tezepelumab Compared With Placebo in Children 5 to < 12 Years Old With Severe Asthma **Short Title/Acronym:** HORIZON **Protocol Number:** D5180C00016 **NCT Number:** NCT06023589 **Sponsor Name:** AstraZeneca (in collaboration with Amgen) **Study Status:** Recruiting **Investigational Product:** Tezepelumab (Trade Name: Tezspire; ATC Code: R03DX11) **Phase of Development:** Phase 3 **Medical Monitor:** Asa Rosqvist / AstraZeneca Clinical Operations Lead

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2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the efficacy (specifically the effect on severe asthma exacerbations) and safety of tezepelumab compared with placebo in children 5 to < 12 years old with severe uncontrolled asthma on medium to high-dose inhaled corticosteroids (ICS) and at least one additional asthma controller medication with or without oral corticosteroids. **Secondary Objectives:** To evaluate the long-term safety, lung function, and asthma symptom control improvements resulting from tezepelumab administration in this pediatric population.

2.2 Background & Rationale Asthma is a chronic respiratory disease manifested as difficulty breathing due to the narrowing of bronchial passageways. An estimated 5% to 10% of pediatric asthmatics have severe symptoms despite adherence to current standard-of-care therapies (medium-to-high dose ICS plus controllers). These asthma exacerbations can be life-threatening, significantly impact the participant's quality of life, and result in significant healthcare costs. There is a critical medical necessity to develop and evaluate novel biologic therapies, such as tezepelumab, to effectively reduce exacerbations and improve clinical outcomes in this vulnerable pediatric demographic.

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3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 2:1 (Tezepelumab : Placebo), Parallel-Group

3.2 Planned Enrollment & Duration Target N: 231 evaluable pediatric participants **Number of Sites:** Multicenter (Global) **Estimated Study Start:** July 2023 **Estimated Primary Completion:** May 2028

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4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Written informed consent from at least one parent/caregiver (as per local guidelines) and accompanying informed assent from the participant (where able) prior to admission to the study. 2. Children 5 to < 12 years of age at the time of signing the assent/consent and at Visit 3. 3. Documented physician diagnosis of severe asthma confirmed and evaluated for at least 6 months prior to Visit 1. 4. Documented physician-prescribed treatment with a total daily dose of either medium or high dose inhaled corticosteroids (ICS) for at least 3 months with a stable dose ≥ 1 month prior to Visit 1. 5. Documented treatment with at least one additional maintenance asthma controller medication (e.g., long-acting beta agonist, leukotriene receptor antagonist, long-acting muscarinic antagonist) for at least 3 months with a stable dose ≥ 1 month prior to Visit 1. 6. Supportive evidence of asthma as documented by at least one of the following: * Post-BD (albuterol/salbutamol) responsiveness of FEV1 $\ge 10\%$ * Positive methacholine challenge (PC20 ≤ 16 mg/mL) * PEF average daily diurnal variability $> 13\%$ * Variability of FEV1 $\ge 12\%$ between clinical visits * Positive exercise challenge test (fall in FEV1 $> 12\%$) * FeNO ≥ 20 ppb despite confirmed ICS maintenance therapy. 7. History of at least 2 severe asthma exacerbation events OR 1 severe asthma exacerbation event resulting in hospitalization within 12 months prior to Visit 1. 8. Pre-BD FEV1 $>50\%$ and $\le 95\%$ PN OR FEV1/forced vital capacity (FVC) ratio ≤ 0.85 at Visit 1 or Visit 2. 9. Evidence of uncontrolled asthma, with at least 1 of the following: ACQ-IA score ≥ 1.5, use of reliever medication on 3 or more days per week, sleep awakening due to asthma symptoms, or asthma symptoms 3 or more days per week during Screening/Run-in. 10. Body weight ≥ 16 kg at Visit 1 (Screening) and Visit 3 (Randomization).

4.2 Exclusion Criteria 1. History of vocal cord dysfunction, cystic fibrosis, primary ciliary dyskinesia, or chronic rhinosinusitis with nasal polyposis. 2. History of any clinically significant disease or disorder other than asthma which may put the participant at risk or influence the results of the study. 3. History of a clinically significant deterioration in asthma or asthma exacerbation including those requiring use of systemic corticosteroids or increase in the maintenance dose of oral corticosteroids within 30 days prior to Visit 1. 4. Change in ICS dose within 1 month prior to Visit 1. 5. History of a life-threatening asthma exacerbation resulting in a hypoxic seizure or requiring intubation.

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5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Tezepelumab or matching Placebo administered via subcutaneous (SC) injections. **Route of Administration:** Subcutaneous (SC) **Duration of Treatment:** 52-week double-blind Treatment period, optionally followed by a 104-week open-label Active Treatment Extension period, and a subsequent 12-week post-treatment follow-up.

5.2 Concomitant & Prohibited Therapy Permitted: Maintenance background asthma controller medications (medium/high-dose ICS and at least one additional controller like LABA/LAMA/LTRA) with or without oral corticosteroids. **Prohibited:** Changes to the baseline ICS/maintenance dose within 1 month prior to screening and concurrent use of other investigational biologics or immunosuppressive therapies.

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6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :--- | :--- | :--- | | **Screening/Run-in** | 4 to 6 weeks | Informed Consent/Assent, Medical History, Baseline Lung Function (FEV1, BD responsiveness), Symptom Evaluation | | **Baseline (Visit 3)**| Day 0 | Randomization (2:1), First SC Injection, Diary Dispensation | | **Interim** | Up to Week 52 | Double-blind SC injections, Safety Labs, Efficacy/Exacerbation Assessments, Pulmonary Function Tests | | **Extension & End**| Week 52 to 168| Entry into optional 104-week open-label extension; Final 12-week off-treatment Follow-up |

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7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Annualized severe asthma exacerbation rate (AAER) over the 52-week double-blind treatment period. **Measurement Method:** Clinical evaluation and documentation to assess the effect of tezepelumab on severe asthma exacerbations in children 5 to < 12 years old with severe uncontrolled asthma compared with placebo.

7.2 Secondary & Exploratory Endpoints * Change from baseline in pre-bronchodilator (pre-BD) forced expiratory volume in 1 second (FEV1) percent predicted normal (PN). * Annualized severe asthma exacerbation rate (AAER) associated with allergic asthma. * Time to first severe asthma exacerbation. * Safety and tolerability profile (Adverse Events and Serious Adverse Events).

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8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring via caregiver diaries, HCP interviews, and clinical site visits throughout the 52-week DB period and the 104-week open-label extension. **Serious Adverse Events (SAEs):** Standard reporting timeline within 24 hours of investigator awareness, with a specific focus on life-threatening exacerbations and hypersensitivity. **Data Monitoring Committee (DMC):** External/Internal independent monitoring committee organized by AstraZeneca to oversee pediatric safety.

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9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy analysis; Safety Set for all participants receiving at least one dose of the investigational product. **Sample Size Justification:** Target $N = 231$, randomized 2:1 to provide sufficient statistical power ($\alpha = 0.05$) to detect a significant reduction in the annualized asthma exacerbation rate. **Handling of Missing Data:** Standard multiple imputation or negative binomial regression methodologies to account for early withdrawals or missing diary data.

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10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits to assure the safety of the pediatric population and adherence to the protocol. **Audit Trail:** Complete electronic Case Report Form (eCRF) tracking and data clarification processes for caregiver-reported symptom diaries and clinical exacerbation inputs.

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11. Study Identifiers & Governance

Primary ID: D5180C00016 **Registry Number:** NCT06023589 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** HORIZON **Official Regulatory Title:** A Multicentre, Randomised, Double-Blind, Parallel-Group Placebo-Controlled, Phase 3, Efficacy and Safety Study of Tezepelumab in 5 to < 12 Year Old Children With Severe Uncontrolled Asthma

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12. Clinical Framework (Asthma Specific)

Primary Condition: Severe Uncontrolled Asthma (Severe asthma in children) **Diagnosis Threshold:** Documented severe asthma ≥ 6 months with ≥ 2 severe exacerbations/year (or 1 hospitalized event) despite standard therapy. **Medical Methodology:** Add-on biological therapy alongside Guideline Determined Medical Therapy (ICS + controller). **Optimization Target:** Reduction in the frequency of severe exacerbations and improvement in lung mechanics.

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13. Study Procedures & Methodology

Management Pathway: Stepwise pediatric asthma management framework incorporating SC biologic administration. **Education Protocol (HCP):** Site training on pediatric SC injection protocols, biological handling, and capturing severe exacerbation events. **Education Protocol (Patient):** Caregiver and patient instruction on recognizing asthma symptom deterioration, using symptom diaries/eDiaries, and handling concomitant rescue medications. **Quality Audit Mechanism:** Tracking site compliance with randomization, drug accountability, and timely entry of exacerbation events into the eCRF.

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14. Observation & Follow-Up Schedule

Total Duration: Up to 174 weeks (4-6 weeks run-in + 52 weeks DB treatment + 104 weeks open-label extension + 12 weeks follow-up). **Onsite Visit Cadence:** Regular intervals aligned with the SC injection dosing schedule. **Remote Monitoring Frequency:** Intermittent remote check-ins for diary compliance and exacerbation monitoring. **Checklist Review Interval:** Regular review of the daily asthma symptom and rescue medication diary.

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15. Sign-off & Approval

Role	Name	Signature	Date
Principal Investigator	[Name]	____	[DD-MMM-YYYY]
Clinical Operations Lead	[Name]	___	[DD-MMM-YYYY]
Lead Statistician	[Name]	_____	[DD-MMM-YYYY]